

José R. Sousa

Charlotte, NC 28217

LinkedIn: <https://www.linkedin.com/in/jrsousa2/>

Phone: 904-290-2890

E-mail: jrsousa2@gmail.com

GitHub: <https://jrsousa2.github.io/>

SUMMARY OF QUALIFICATIONS

- Extensive experience with **databases**, **data warehouses** and **SQL**.
- Strong **Python** skills using **VS Code**. Ran **SQL** queries, made **web API** requests to retrieve data, created **GUI's**, developed **REST API's**, loaded **Excel** and **XML** files into **pandas** dataframes, read and wrote to Excel files with **openpyxl**, created formatted **PowerPoint** slides automatically, created **virtual environments**, automated **batch jobs**, etc. [All can be viewed on my GitHub](#).
- Extracted data from **Hadoop** with Dremio / Hive and created VDS's.
- Fitted **logistic regression** models using both **SAS** and **PySpark ML**.
- Independently created **logistic regression** model in **SAS** to score risk of auto policyholders. The model was deployed.
- Created project on fitting a logistic regression with **PySpark** to a public credit risk dataset. That involves ETL to ingest, clean, transform, and prepare the data before modeling.
- Worked on **insurance pricing models** in partnership with consultancies.
- **Expert at SAS** (**Enterprise Guide** and **Viya**) -- created **stored processes** and **ETL's**, wrote advanced code to produce tables and reports, and truly mastered **SAS macros**. Created dashboard with SAS/GRAPH and reports with **Proc Report**.
- Strong with **Machine Learning** models and **data analytics** and familiar with **actuarial analyses** and **actuarial reports**.
- Expert at **MS Excel**, created advanced spreadsheets with **VBA macros**, **automated charts** and **graphs**.
- Created **workflows** in **n8n** for Gen AI.
- Built a workflow in n8n that extracts content, transforms it with Python scripts (JSON/HTML processing, metadata generation, etc.), and loads it into downstream publishing steps. Worked on n8n workflow JSON and HTML documentation, where Python manipulated structured data before producing the final artifacts.
- Built an alert code in Python and scheduled it on GitHub (**serverless** process).
- Most of my experience is in the **insurance** industry (15+ years), P&C, personal and commercial. Created a wide variety of **loss triangles**, estimated **ultimate claims** from **LDF's** (**reserving analysis**), calculated ceded losses from **reinsurance** contracts, calculated earned premiums, created **loss ratio reports** and **in-force policies**, capped **GC** losses, performed **reconciliation** with the GL, etc.
- Exposure to **accounting principles** – using the **GL** Oracle database, created **YTD summary reports** with array formulas in **Excel** to support quarterly filings (10-Q and 10-K) and performed account and profit center level reconciliations with it.
- Graduated in **Math / Statistics** from the **best university in Brazil** (**U of São Paulo**).
- Very analytical, self-taught, can **work independently**, created new formulas in math (e.g. <https://arxiv.org/html/1810.07877v7>).
- **U.S. citizen** (had a **green card** through **NatGen** previously).

SOFTWARE

- **Databases:** **Oracle**, **Teradata**, **SQL Server / SSMS** and their **IDE's**.
- **Big Data:** **Snowflake**, **Databricks**, **PySpark**, **Hadoop**, **Dremio**, **Hive**.
- **Programming:** **Python**, **VBA**, **VBScript**, **Unix/Windows batch scripting**.
- **Cloud:** Working knowledge of **AWS**, **Azure** and **GCP**.
- **SAS:**
 - Platforms: **Unix**, **Linux**, **Windows Server** and **Viya** (Cloud).

- Modules: BASE, STAT, GRAPH, ACCESS, CONNECT.
- Expert knowledge of most used SAS Procs, including **Proc Report**. Advanced **macro language** and **stored processes**.
- **BI / Predictive Modeling**: SAS **Enterprise Guide**, **Power BI**.
- **Code Editors**: **VS Code**, Jupyter Notebook.
- **Other**: **Excel**, **PowerPoint**, SharePoint, **Git**, **RegEx**, **Jira**, **XML**, **JSON**.

PROFESSIONAL EXPERIENCE

Wells Fargo (**banking**)

Senior Data Analyst (**contract** / **remote** / **hybrid**)

10/2022 – 12/2025

- **Second project (hybrid)**
 - Worked on **AML** / **Financial Crimes** remediations (CIP / BO / CDD / **KYC**).
 - Built AML remediation populations (Trust, NRM4 and transportation assets).
 - Created population identification workbooks and waterfall charts and presented them to senior directors for approval.
 - Wrote complex **optimized SQL** queries to extract data from **Hadoop** using **Dremio** and **Hive** and from various databases (**Teradata**, **Oracle**, SQL Server, DB2, etc.). Was a go-to person for **challenging** technical questions.
 - Developed **SAS** and **SQL** code to create ad-hoc, time-sensitive data extracts and **reports**.
 - Created code test scripts and worked with code testers to validate the code.
- **First project (remote)**
 - Worked on **customer remediations**, responsible for the creation of the final artifacts from which refunds and letters are issued to customers as required by **regulators**, following strict **Confluence** procedures.
 - Managed **SAS** code used to create final remediation artifacts, made changes to the code to add **enhancements**, **optimize performance** and minimize the number of **defects** and findings in **peer reviews**. As an example of enhancement, when a new analyst takes over a remediation, all the **artifacts** and **SAS** tables created by the previous analyst are **automatically** saved to a new subfolder as **backup**.
 - Created **ETL**'s to produce new artifacts by mapping fields from the execution data requirements (EDR) to data tables. Contributed to design **artifact layouts**.
 - Used **GitHub** for code version control and sharing and **DMS** as a secure repository for storing and sharing artifacts.

Truist, Charlotte, NC (**banking**)

Data Analytics SQL/Python (**contract** / **remote**)

09/2021 – 03/2022

- Created rules based **decision tool** for the compliance and data analysts to choose their optimal analytics application as a web form created in **Python**.
- Created and maintained current report on user accesses for the various **proprietary applications** and data repositories (**AD**, **ACL**), for **compliance**.
- Ran reconciliation between **ServiceNow**/RITM and data pulled from the **mainframe** to assess correctness of an automatic ticket creation process for abends (abnormal job termination), and presented findings with **PowerPoint**.
- Ran **SQL** queries in **Snowflake** to extract data from the **data lake/data warehouse**.
- Created reports using the **SAS grid** and produced **documentation** following internal **templates**.

Florida Blue, Jacksonville, FL (**healthcare**)

SAS/Python Developer (**contract** / **remote**)

04/2020 – 08/2021

- Developed **ETL** code to extract data from **Oracle** and **DB2** and others, using **SAS 9.4** on **AIX** (Unix) and **Enterprise Guide**.
- Extracted data from multiple sources to create agent **fraud detection** analytics for the **SIU** dept with various metrics. Created **macro to automate** the formatting of the report for each metric using **Proc Report**.
- Automated and scheduled various **SAS ETL** jobs in **Unix**, using a **SAS macro**.
- Used **HTML** syntax in **SAS** to compose rich text e-mail alerts.
- Combined Windows **batch**, **WinScp** and **VB** scripts to schedule jobs and automate file transfers and moving data between platforms using SFTP.
- Created a neat **SAS stored process** for various canned reports.
- Created **visualizations** and **reports** using **Power BI** and **Python**, using pre-modeled datasets (no need for DAX customization).
- Gave **training** to team members on how to design **SAS stored processes**, covering how to create the **GUI** and how to handle the variable **user's inputs**.

National General (Allstate), New York, NY (insurance)

Insurance Data Programmer / Data Manager

04/2015 – 12/2019

- **Reported to the holding company's chief actuary.**
- Mastered the **claims data system** of the company (EPIC) and legacy data stored in **Oracle** and **SQL Server** databases.
- **Planned** and **created** datasets in **SAS** by extracting and combining data from various databases, from which loss triangles and **data analytics** were produced.
- Created **actuarial loss triangles** used in reserving analyses with **SAS macros**, and **reconciled** the numbers with the company's **general ledger**.
- Created quarterly company-wide **YTD summary report** in **Excel** using **general ledger data** extracted with **Oracle BI Discoverer** to support quarterly filings and the annual report preparation (**10K**, **10Q**), for the **SEC** and other **regulators**.
- Participated in the **ATLAS project**, with a goal to capture claim and policy level data into the **GL Oracle database**.
- Extracted data from the **SAS server** to create reports with **SAS macros** and **code**. These reports were used internally or provided to consultants, **auditors** and **regulators**.
- Using **SAS**, ran **statistical data analyses** to track performance of products.
- Created complete documentation about work done on legacy data from acquired companies.

AIG, New York, NY (insurance)

Data Scientist / SAS Programmer (contractor / FTE)

12/2011 – 12/2014

- Planned and extracted data from multiple sources (**mainframe**, **DB2**, **SQL Server** and **Oracle**) to produce **enhanced tables** used in all **reports** and **models** and **processes**, using **ETL** and **SAS/Access** interface to **OLE DB** and **ODBC**.
- Planned the layout and created enhanced tables from which loss triangles, actuarial reports, predictive models (**AQI**) and other processes were fed.
- Created **ETL's** to read **VB/VSAM** files on **IBM mainframe** and load them into **SAS** tables on Windows. These files became the data foundation of the **Global Actuarial Pricing** department.
- **Reconciled** quarterly actuarial data loads created from ETL's with control totals.
- Wrote **SAS macro** to create a wide variety of **loss triangles** used to derive **LDF's** and **estimate** the **ultimate claims**.
- Worked in the **deep dive** and **profitability analyses** of the book of business, by extracting data on **losses**, **premium**, **policy** and **insured attributes** to generate various reports and metrics and to monitor the **performance** of the various **lines of business and segments**. Performed claim analytics and policy analytics.

- Created smart **algorithm for linking policies** based on insured and policy characteristics in **SAS**. This policy linking was the basis for the creation of a **scoring model** called AQI (**Account Quality Index**).
- Worked with Deloitte consultants and provided them with **loss triangles** used as inputs into the Commercial Auto **pricing tool**. Also worked with Deloitte on the **Specialty Worker's Comp model**, making sure that deliverables such as **SAS code**, **data** and results of the **predictive model** met the requirements.
- Created and scheduled weekly **jobs** to extract submission data entered by underwriters into the **AQI model** and update a **Qlikview dashboard**.

Capital One, Toronto, ON (banking)

- *Data Analyst (contractor)*

01/2011 – 11/2011

- Worked on various AML efforts (KYC / CDD), using **Excel/VBA** and **PowerPoint** as main reporting tools.
- Created advanced dashboard with **SAS/Graph** which ran weekly on Unix and was sent to a distribution list to monitor web performance.
- Tweaked and ran some regular critical processes of the company, such as, file transfers to vendors and others, via **Unix scripting** (ksh, sh, SQL).
- Created ETL jobs and scheduled them to run in Unix to load daily files received from vendors into **SAS** and **Teradata**.
- Created Excel workbook with VBA automation to retrieve data from tables in Teradata and copy formula results from this workbook into text boxes and graphs in a **PowerPoint** deck used in the **AML** monitoring process.
- Automated customer bookings report with SAS **Proc Report**.

Aviva Canada, Toronto, ON (insurance)

- *Senior Data Analyst (contractor / FTE)*

05/2008 – 12/2010

- Independently created logic and **SAS code** to extract **in-force policies** from the commercial lines book of business.
- Independently designed **SAS stored processes** to automate the creation of **loss ratio reports** with a straightforward user-friendly GUI.
- Planned and provided data used by EEA consultants to develop risk and pricing models, with accurate match between policies and claims.
- Planned, wrote and ran **SAS** code used to **reconcile** monthly actuarial data loads.
- Created extensive **documentation** on all existing data and processes via **HTML** and **VBA**, such as table layouts, variables description and distribution, etc.
- Investigated and troubleshooted a lot of data problems reported by the users.
- Created facts and dimensions tables with SAS **OLAP** Cube Studio.
- Tweaked and updated monthly actuarial reports with **BusinessObjects**.
- With support from the actuaries, designed and coded the Fleet Rate Tracking Report in **SAS**, a project that required rates to be deconstructed with the use of **VBA** coding.

EDUCATION

Master's in **Statistics** - **University of São Paulo, Brazil**

Bachelor's in **Statistics** - **University of São Paulo, Brazil**

Bachelor's in **Mathematics** - **University of São Paulo, Brazil**